ELSEVIER

Journal of Medical Imaging and Radiation Sciences xx (2019) 1-12

Journal of Medical Imaging and Radiation Sciences

Journal de l'imagerie médicale et des sciences de la radiation

www.elsevier.com/locate/jmir

Review Article

Machine Learning Methods for Computer-Aided Breast Cancer Diagnosis Using Histopathology: A Narrative Review

Shweta Saxena, MTech* and Manasi Gyanchandani, PhD

Maulana Azad National Institute of Technology, Bhopal, Madhya Pradesh, India

ABSTRACT

Histopathology is a method used for breast cancer diagnosis. Machine learning (ML) methods have achieved success for supervised learning tasks in the medical domain. In this article, we investigate the impact of ML for the diagnosis of breast cancer using histopathology images of conventional photomicroscopy. Cancer diagnosis is the identification of images as cancer or noncancer, and this involves image preprocessing, feature extraction, classification, and performance analysis. In this article, different approaches to perform these necessary steps are reviewed. We find that most ML research for breast cancer diagnosis has been focused on deep learning. Based on inferences from the recent research activities, we discuss how ML methods can benefit conventional microscopy-based breast cancer diagnosis. Finally, we discuss the research gaps of ML approaches for the implementation in a real pathology environment and propose future research guidelines.

RÉSUMÉ

L'histopathologieestuneméthodereconnue de diagnostic du cancer du sein. Les méthodesd'apprentissage machine (AM) ontobtenu un grand succès pour les tâchesd'apprentissagesupervisédans le domainemédical. Danscet article, nous examinons les répercussions de l'AM pour le diagnostic du cancer du sein enutilisant des images d'histopathologie de photomicroscopieconventionnelle. Puisque le diagnostic de cancer porte sur l'identification de l'imagecommecancéreuseou non-cancéreuse, il suppose le prétraitement de l'image, l'extraction des caractéristiques, la classification et l'analyse du rendement. Danscet article, nous examinonsdifférentesapproches de l'exécution de cesdifférentesétapesnécessaires. Nous avonsconstaté que la plupart des recherches AP pour le diagnostic de cancer du sein ontmis l'accent sur l'apprentissageprofond. Sur la base des inférencestiréesd'activités de recherchesrécentes, nous discutons de la façondont les méthodesd'AMpourraientprofiter au diagnostic du cancer basé sur la microscopieconventionnelle. Enfin, nous discutons des lacunes de recherche des approchesd'AM pour la miseenœuvredans un environnementréel de pathologie et proposons des lignesdirectrices pour des recherches futures.

Keywords: Breast cancer; machine learning; computer-aided diagnosis; histopathology

Introduction

Breast cancer may occur because of abnormal cell division in the breast, which develops a mass called a tumor. There are two types of tumors: cancerous (malignant) and noncancerous (benign) [1]. The most commonly used techniques for cancer screening and diagnosis are mammography, magnetic resonance imaging, fine needle aspiration cytology (FNAC), and histopathology [2]. Among these, histopathology is recognized as the gold standard for cancer diagnosis [3]. In this method, infected tissue of the breast is removed and examined under the microscope. However, the histopathology slide contains intricate visual patterns that are difficult to categorize as benign or malignant [4]. Computer-aided diagnosis (CAD) system quickly analyzes the images and provides a diagnostic opinion, which helps to enhance the accuracy of pathological decisions [5].

Machine learning (ML) models for cancer diagnosis have been popular since the advancement in digital microscopy. In 1993, an ML-based CAD model was developed by Street et al [6]. It was first used at the University of Wisconsin and soon became available for remote execution via the World

Contributors: All authors contributed to the conception or design of the work, the acquisition, analysis, or interpretation of the data. All authors were involved in drafting and commenting on the paper and have approved the final version. Funding: This study did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Competing interests: All authors declare: no financial relationships with any organizations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

* Corresponding author: Shweta Saxena, MTech, Maulana Azad National Institute of Technology, Bhopal, Madhya Pradesh, India.

E-mail address: shwetasaxena631@gmail.com (S. Saxena).

1939-8654/$ - see front matter
https://doi.org/10.1016/j.jmir.2019.11.001

Wide Web. Furthermore, the researchers developed a breast cancer data set using FNAC, and made it freely available on the UCI ML repository [7]. Since the development of this model, several researchers have been trying to develop different CAD systems for various applications such as breast cancer, lung cancer, skin cancer, prostate cancer, brain cancer, colonial cancer, cervical cancer, bladder cancer, and liver cancer [8–11]. However, the development of an accurate, reliable, and fast CAD system for a real pathology environment is still a challenging task. There are four steps in performing the cancer diagnosis using a ML-based, CAD system: preprocessing, feature extraction and selection, classification, and performance analysis [2,12]. In the past few years, several researchers provided reviews of state-of-the-art methods for developing a CAD system. For example, in the study by Gandomkar et al [13], the authors discussed methods for grading, classification, and segmentation of whole slide images of breast cancer. A review of different stages for developing CAD systems for mammography, magnetic resonance imaging, FNAC, and histopathology is provided in the study by Jalalian et al [2]. The article by Komura and Ishikawa [10] focuses on ML methods for the analysis of whole-slide histopathology images. In the study by Aswathy and Jagannath [14], the researchers focused on segmentation-based approaches for the diagnosis of breast cancer using histopathology. The literature [15] covers deep learning for medical diagnosis in general. The article by Huet al [11] covers deep learning methods for the image-based detection and diagnosis of cancer. Most of the discussed review articles focused on whole-slide image–based cancer diagnosis and consider segmentation as an essential step of preprocessing. However, the scope of the whole-slide scanner is limited in a real pathology environment because of the high implementation cost and operating technology [16]. Photomicroscope is a commonly used instrument for the examination of histopathology images for cancer diagnosis in real pathology environments [17]. This article is focused on reviewing ML methods for breast cancer diagnosis using conventional photomicroscopy-based histopathology images.

The remaining portion of this article is structured as follows: Sections Preprocessing, Feature Extraction, Dimension Reduction, Classification, Performance Evaluation are focused on discussing the necessary steps of the ML-based CAD model. Each section introduces the step and discusses the ML method to implement the step. Comparison of Published Results provides a comparison of published results. The Discussion outlines the impact of using ML for histopathology and research gaps for the application of these methods in a real-world environment. Finally, the Conclusion provides future research guidelines.

Preprocessing

Preprocessing is required to convert the image into a simpler and effective form [18]. It reduces the noise in the image and helps to identify the focal areas [19]. Contrast-limited adaptive histogram equalization approach is used to improve the local contrast [20]. Thresholding is used to reduce the noise present in the image. Pixels under a threshold value in the intensity histogram of an image are considered as noise [19]. The optimal threshold is calculated by Ostu's threshold method [21]. Background correction and filtering are used to improve the results of thresholding. Background correction minimizes the effect of different lighting conditions, and filter reduces the random noise present in the image [19]. The approaches mentioned previously are commonly used to enhance the quality of digital images for ML applications. For deep learning–based CAD systems, the most commonly used preprocessing strategies are data augmentation and color normalization. In data augmentation, the training data set is enhanced using various image transformation methods such as scaling, flipping, mirroring, blurring, and adding noise. These transformations modify the image morphology [22]. Moreover, histopathology images vary in color and illumination because of the discrepancies in the preparation of histopathology slides (stain coloring) and different lightning conditions while capturing the digital image. Therefore, an essential step in histopathology image preprocessing is stain color normalization [14]. In this method, a histopathology image with a perfect stain coloring is selected as the reference image. The color values are adjusted so that it matches the color values of the target image. A research study [23] shows that stain normalization enhances the performance of deep learning–based classifiers. The researchers in [24,25] used stain normalization for developing a deep learning–based CAD system for breast cancer histopathology images. A comparison of recently used stain color normalization methods for histopathology images is presented in the study by Roy et al [26]. Some research studies proposed very effective methods of color normalization in digital images. For example, the researchers in [27] proposed a color normalization method that converts the RGB image into perception-based color space ($l\alpha\beta$). The method transfers the color between the source and the target image by calculating the mean and standard deviation for each axis of the image. The output color channel (O) can be calculated by using Equation 1:

$$O = \left[\left(\frac{S - \mu(S)}{\sigma(S)} \right) * \sigma(T) \right] + \mu(T) \tag{1}$$

Here, S is the pixel intensity of the source channel, μ (S) is the mean value of the source channel, σ (S) is the standard deviation of the source channel, σ (T) is the standard deviation of the target channel, and μ (T) is the mean value of the target channel. The researchers in [28] proposed the method for color normalization of histopathology images based on quantitative analysis. In the study by Khan et al [29], the authors proposed a stain normalization method that computes image-specific stain matrix using stain color descriptor. After that, different stain concentration values of the image are obtained using color deconvolution (stain separation) and provided to the nonlinear (spline based) mapping function. Finally, the source image is recreated using normalized stain channels. A recent study [22] shows that stain normalization should be used along with color augmentation for obtaining the best performance of deep learning–based CAD systems.

Feature Extraction

After preprocessing, the images are converted into a unique feature vectors [30]. Feature extraction plays a vital role in designing an ML model [31]. Feature extraction methods found in the literature for CAD can be classified into two categories: handcrafted (manually designed) features and deep learning–based features. In designing handcrafted features, a user has to put many efforts to decide which features are relevant for the diagnosis. By contrast, deep learning models automatically determine the essential features while training the network for the classification. This section provides a review of feature extraction methods used for CAD of cancer in the past few years. Handcrafted feature extraction methods and their limitations for breast cancer diagnosis using histopathology images are discussed in section Handcrafted Feature Extraction Methods and Their Limitations. Section Deep Learning–Based Feature Extraction Methods and Their Limitations discusses deep learning based feature extraction methods and their limitations for the diagnosis of breast cancer using histopathology.

Handcrafted Feature Extraction Methods and Their Limitations

A CAD system based on spatio-color- texture graph segmentation, texture feature extraction (GLCM, GRLM, and Euler number), and linear discriminant analysis–based classification was proposed by the authors in [32]. The system classified 70 histopathological images of the breast with 100% accuracy. In the study by Vu et al [33], the discriminative feature–oriented dictionary learning method obtained 97.75% accuracy for data sets of human intraductal breast lesions and Animal Diagnostic laboratory. A CAD system based on morphological feature extraction achieved 85.7% accuracy in classifying histopathological data set of 70 images [5]. However, a significant limitation of these research studies was the small unpublished data set for the experiments. Owing to this limitation, further comparison of these works with the other studies is challenging. To overcome this restriction, the authors in [3] introduced a publicly available breast cancer histopathological data set (“BreaKHis”) and performed some initial experiments using handcrafted features. The authors observed that the PFTAS features deliver the highest cancer recognition rate as compared with the LBP, CLBP, LPQ, ORB, and GLCM features. In another study, the Gabor filter, wavelet features, and LBP features along with an ensemble classifier scored 90.32% accuracy in classifying the images of BreaKHis on 200× magnification [34]. Recently, PFTAS features along with a multiple-instance learning–based nonparametric classifier achieved the highest patient recognition rate (Prr) for BreaKHis data set [35]. However, this research undergoes a significant standard deviation in accuracy.

Deep Learning–Based Feature Extraction Methods and Their Limitations

Deep learning–based feature extraction has gained a remarkable interest among researchers for the classification of histopathology images. The research study [36] shows that deep learning–based features attained higher accuracy as compared with color, texture, and geometrical features for the automatic detection of invasive ductal carcinoma tissues in the whole-slide images of breast cancer. The authors in [3] have performed some experiments on breast cancer histopathological data set (“BreaKHis”) using LBP, CLBP, LPQ, ORB, GLCM, and PFTAS features. Later, the same authors in [37] have implemented a deep learning–based model for the classification of the BreaKHis data set. The model achieved higher accuracy compared with the classification models based on LBP, CLBP, LPQ, ORB, and GLCM features. Recently, multiple-instance learning–based convolutional neural network (CNN) achieved the highest Prr for BreaKHis data set [35].

The studies mentioned previously are based on training the CNN model from scratch. However, this kind of training requires a large data set to improve training performance. However, it is difficult to obtain such a large data set from local hospitals because of factors such as associations with pathology laboratories, patient data access permissions, and diagnostic result authorizations. Despite these difficulties, the research study [3] introduced a breast cancer histopathological data set named BreaKHis and made it publicly available for further research. But, this data set is comparatively small and provides limited samples for training in comparison with the large data sets such as ImageNet [38]. Moreover, training from scratch requires many efforts to fine-tune the parameters of CNN and long training time in comparison with the pretrained CNN model.

Histopathology images share similar features such as lines, edges, and colors with the images of ImageNet data set. So, the CNNs previously trained on ImageNet may be utilized for extracting the feature of histopathological images [10]. Therefore, the researchers are motivated to use pretrained CNNs in place of the CNNs trained from scratch. A recent study [39] demonstrates that pretrained CNNs are good substitutes for the CNNs trained from scratch for the diagnosis of breast cancer using histopathology. This work uses transfer learning to develop CAD models for breast cancer. The limitations of this research are delineated in the following.

1. It does not give a description of how the input image is provided as input to the pretrained CNNs. It seems that they have resized the image for giving it as input to the pretrained VGG16, VGG19, and ResNet50 for feature extraction.
2. It is not exhaustive as it considers the pretrained VGG16, VGG19, and ResNet50 only for the feature extraction.

3. The experimental setup is not same as [3] and the work is not compared with [3,37,40].

In some studies [24,41], the pretrained CNN was used to classify the BreaKHis data set. These works have used different combinations of training and testing folds, so their performance cannot be compared with [3,37,40]. These works have also resized the input image to use pretrained CNNs, which may lead to loss of information. In the study by Spanhol et al [40], pretrained BVLC CaffeNet (modified AlexNet) was used to extract deep features (DeCAF) and logistic regression was used for classification. The accuracy reported by this work was less than that of the combination of CNNs and multiple-instance learning–based CNN trained from scratch [35,37]. In addition, the features of other pretrained CNNs such as AlexNet [42], VGGNet [43], GoogLeNet [44], and ResNet [45] has not been explored and compared with the features proposed in this study.

Dimension Reduction

Feature extraction generally gives a large dimensional feature vector corresponding to an image. Many of the features in a feature vector could be irrelevant or redundant. If a large number of features are provided into the classifier for diagnosis of disease, then the classifier takes a high computational cost and time to classify the features. Moreover, if we have the large dimensional feature vectors and a little training data (as in case of medical diagnosis), then it may cause overfitting the training data set and the ML model may not recognize the unseen images as compared with the training images [46]. Dimension reduction is a commonly used approach to improve the computational complexity of a classifier on the cost of slightly sacrificing the overall performance of the classifier [47]. There are two ways to perform dimension reduction [48]:

1. Selecting the essential features from original dimensions (feature selection).
2. Constructing new dimensions.

Selecting the Relevant and Important Features from Original Dimensions

In this method, different feature combinations are formed and evaluated to obtain the best combination. Some heuristic algorithms for feature selection are sequential forward selection and sequential backward selection [49]. However, heuristic methods are not guaranteed to be optimal for the selected features but perform well as compared with the other feature selection algorithm [47,50]. Some recently used feature selection schemes are genetic algorithm, simulated annealing, boosting [51], grafting [52], and particle swarm optimization [53].

Constructing New Dimensions

This method maps the original feature space into a new feature space of reduced dimensions. Commonly used dimension reduction techniques are principal component analysis, linear discriminant analysis, independent component analysis, and manifold learning [47].

Limitations of Dimension Reduction and Possible Solution

Although dimension reduction improves the computational complexity of a classifier, it may cause sacrificing the loss of essential features that may help in predictions. So, the overall performance of the classifier may be compromised because of dimension reduction. Deep learning–based models have several pooling layers to reduce the dimensions of image feature maps [54]. When the image feature maps are converted to the single-dimensional feature vector, a very high dimensional feature vector is generated for the classification. For a large dimensional feature vector, it is challenging to decide which features are relevant for the diagnosis and which are not. So, in place of dimension reduction, an alternate method called dropout [55] can be used to solve the overfitting problem in deep learning. Another method to avoid overfitting in ML is regularization [56]. In this method, during learning, each feature's weight is penalized by using the regularization parameter λ. The value of λ should be carefully selected by the user. Increasing the value of λ significantly reduces the chances of overfitting. However, if λ is increased beyond a limit, it may cause underfitting, that is, the model may not be classifying the training data correctly which may lead to a significant training error [57].

Classification

Classification is a process of dividing the data into two or more classes based on quantitative information inherent in the feature vector [58]. For breast cancer diagnosis, data points need to be classified into two groups—benign and malignant. The classification model consists of two phases—training and testing. With training, feature vectors containing the class labels are provided as input to the classifier. These feature vectors can be considered as learning examples for the classifier. Once a classifier is trained on examples, it can be used for testing a feature vector of an unknown class. The outcome of testing is a class label corresponding to a feature vector. ML techniques that have been widely applied for the development of CAD models are decision trees, Naïve Bayes, k-nearest neighbor, artificial neural network (ANN), support vector machines (SVMs) and ensemble classifiers [8,59]. Research study [8] provides the applications, advantages, and limitations of these classifiers for cancer prediction and prognosis.

Decision Tree

It is the "divide and conquer"–based method of data classification. In this method, data can be represented in the form of a tree, where different features are represented by internal nodes, and labels of data samples are represented by leaf nodes. For the classification of the particular data sample, the tree is traversed from root to leaf. The leaf node contains the final classification result [60]. The most commonly used decision tree algorithm for classification is C4.5 [61]. A comparison of C4.5 with some other decision tree algorithms is presented in the study by Lim et al [62]. Salvatore Ruggieri [63] implemented a more efficient version of C4.5 known as EC4.5, which gives 5 times improved performance for the same decision tree as compared with C4.5. It gives the same decision tree as C4.5.

Nearest Neighbor

In this method, a data sample is compared with other data samples using a distance metric. Distance between two data samples can be obtained by measuring the difference between two attribute values. Distance metric should be used such that it minimizes the distance between two similar data samples and maximizes the distance between two dissimilar data samples [30]. Generally, the standard Euclidean distance is used to measure the distance between two data samples. If x and y are two data samples of n-dimensions, then the Euclidean distance between x and y can be given by Equation 2.

$$D(x, y) = \sqrt{\left(\sum_{i=1}^{n} |x_i - y_i|^2\right)} \qquad (2)$$

Data samples which are close to each other are called similar data samples or the nearest neighbors. That is why this method is called the nearest neighbor classification method. To assign a class to a new data sample, its k-nearest neighbors are located, and their class is identified. The majority class will be the resultant class of that data sample. This method is named as k-nearest neighbor method [60].

Artificial Neural Networks

ANN is similar to the biological network of interconnected neurons in the human brain [64]. Most commonly used ANN for classification problem is feed-forward ANN, along with backpropagation training algorithm [65]. The basic structure of a single neuron in a feed-forward ANN is shown by Figure 1 [66]. A particular neuron in ANN receives input information x_i from other neurons, multiplies each datum by corresponding weight w_{ij}, and produces a weighted output $f(x_j)$, with the help of an activation function. This weighted output is again passed as input to another neuron in the next layer, and the same process is repeated until the output layer is reached [67]. Neurons are arranged in the form of layers in this type of network. An input layer, an output layer, and many hidden layers can be present in a typical feed-forward ANN.

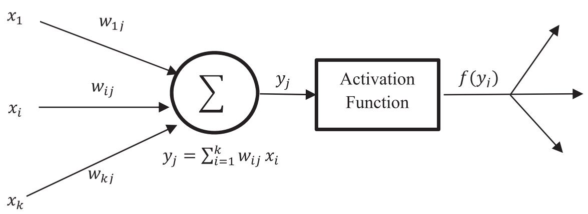

Figure 1. The basic structure of a single neuron in ANN.

The backpropagation training algorithm provides the procedure of changing the weights during training in feed-forward ANN. The actual output produced by ANN in the output layer is compared with the desired output. After this, an error is calculated by finding the difference between the actual (calculated) and desired (target) output. Finally, the error is propagated back into the network for updating weights in the next iteration [68].

Support Vector Machine

SVM maps the input data point (feature vector) into a high-dimensional feature space, and identify a hyper-plane that separates the data points into two classes. An optimal hyperplane can be obtained by maximizing the distances between support vectors (the data points closest to the boundary of the class) of two classes [69]. A gentle introduction of SVM classifier is provided in [70]. SVM can be used to perform the linear and nonlinear classification with the help of the linear and nonlinear kernels. Applying different kernel functions (linear, cubic, quadratic, polynomial, Gaussian) to different histopathology data sets can significantly increase the performance of the SVM classifier [8]. Some of the significant research studies [3,35,71,72] used SVM for the classification of the features of breast cancer histopathology images.

Naïve Bayesian Network

Bayesian network (BN) is a direct acyclic graph, which represents a relationship among a set of variables (features) with the help of probabilities. Figure 2 shows a BN along with the value of conditional probabilities for each variable [30].

Nodes in the graph represent variables and arcs represent relationships among variables [8]. Naïve Bayesian network is a particular case of BN in which DAG contains only one parent and many children. Child nodes remain independent of one another in the context of their parent. BN classifiers are used to produce probability estimations rather than predictions [9].

Ensemble Classifier

With this approach, multiple classifiers are combined to produce more accurate classification results as compared with a single ML classifier. Most commonly used methods to construct an ensemble classifier are bagging, boosting, and random subspace method [59]. Bagging generates a no of sets of training samples from the original data set and trains

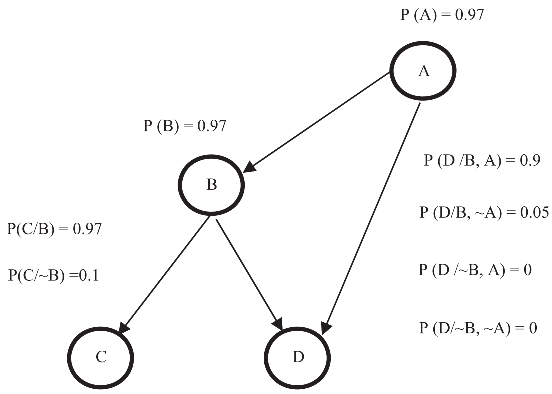

Figure 2. Bayesian network.

a base classifier on these samples [73]. Boosting iteratively builds the classification models, and train those models on a training set. The training set can be modified in each iteration based on the misclassified samples in the previous iteration [74]. In the random subspace classifier, a random feature subset is picked up from the original data set for training each classifier. The outputs of all the classifiers are combined to generate a unique output with the help of the voting scheme [75]. Experimental evidence shows that ensemble methods are more accurate than a single classifier [76].

Convolutional Neural Network

CNN achieved great success in the field of medical image analysis and classification [77–79]. CNN consists of a stack of multiple trainable stages, a supervised classifier, and a set of arrays known as feature maps for representing input and output of each stage [80]. There is no need to design a separate classifier in this method. In the past few years, deep learning–based models, such as AlexNet [42], VGG16 and VGG19 [43], GoogLeNet [44], and ResNet [45] have become very popular for the image classification tasks. The current training strategies found in the literature for cancer diagnosis are training from scratch and transfer learning. Training CNN from scratch considers an input image, apply randomly initialized filters or weights, and pass the convoluted output (features) to the next layer. The weights of initial layers learn the standard features, such as edges and corners, from the image and start learning the task-specific features as the image passes to the deeper layers. The weights are updated in each iteration to minimize the classification error. As soon as the classification error is reduced, learning stops. These weights can be stored in memory and can further be used for initializing the same network for the classification of a different set of images. This approach of reusing the weights of the previously trained network is called transfer learning [39].

Performance Evaluation

In the histopathology data set, multiple histopathology images may correspond to a single patient. Therefore, the performance of the CAD system is measured by the researchers in terms of the image-wise diagnosis and patient-wise diagnosis. For the image-wise diagnosis, the most commonly used matrices are sensitivity, accuracy, specificity, and receiver operating characteristic (ROC) curve. These matrices can be estimated based on the following formulas [5].

$$Sensitivity = \frac{TP}{TP + FN} \times 100 \tag{3}$$

$$Specificity = \frac{TN}{TN + FP} \times 100 \tag{4}$$

$$Accuracy = \frac{TP + TN}{Total\ data\ samples} \times 100 \tag{5}$$

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

$$F - measure = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{8}$$

Here, true positive value represents the number of correctly classified samples of the total malignant samples, and true negative value in the data set represents the number of correctly classified samples over total benign samples in the data set. The number of misclassified samples of the total malignant samples are called false positive, and the number of misclassified samples of the total benign samples are called false negative [2]. The ROC curve is represented by a two-dimensional graph in which the true positive rate (sensitivity) is represented as a function of false positive rate (1-specificity) for different cutoff points. Generally, the area under the ROC curve is considered as a performance metric for the ROC curve. For better performance of the CAD system, the area under the ROC curve should be large [12]. Moreover, the patient-wise diagnosis is performed using Prr that can be calculated as follows [3].

$$Patient\ score(PS) = \frac{Correctly\ classified\ cancer\ images\ of\ the\ patient}{Total\ number\ of\ cancer\ images\ of\ the\ patient} \tag{9}$$

$$Patient\ recognition\ rate(Prr)(\%) = \frac{Sum\ of\ all\ patient\ scores}{Total\ number\ of\ patients} \times 100 \tag{10}$$

Comparison of Published Results

Data Sets

This section describes the conventional photo microscopy-based histopathology data sets commonly used for the diagnosis of breast cancer.

- BisQue [81]: The biosegmentation benchmark data set (BisQue) contains 58 H&E-stained breast histopathology images of size is 896×768. It includes 32 benign and 26 malignant images which can be downloaded from http://bioimage.ucsb.edu/research/biosegmentation.
- BreaKHis [3]: It is a collection of breast biopsy samples of 82 patients. The images were collected through a clinical study from January 2014 to December 2014 by P&D Lab, Brazil. In this data set, the images are taken on different magnifications: 40×, 100×, 200×, and 400×. There are 1995 images (625 benign and 1,370 malignant) of 40× magnification; 2,081 images (644 benign and 1,437 malignant) of 100× magnification; 2,013 images (623 benign and 1,390 malignant) of 200× magnification; and 1820 images (588 benign and 1,232 malignant) of 400× magnification. For multiclassification, the images were further divided into four subcategories of malignant cancer: ductal carcinoma, lobular carcinoma, mucinous carcinoma, and papillary carcinoma; and four subcategories of benign cancer: adenosis, fibroadenoma, phyllodes tumor, and tubular adenoma. This is the most widely used data set by the researchers for CAD of breast cancer using histopathology. The data set is available on http://web.inf.ufpr.br/vri/breast-cancer-database. Figure 3 shows the digital images of the BreakHis data set perceived on different magnifications.
- Bioimaging 2015 breast histology classification challenge data set [71]: This data set is composed of 269 images of H&E-stained breast histology. The image size is 2048 × 1536 pixels acquired on 200× magnification. For binary classification, there are two categories noncancerous and cancerous. There are two categories of noncancerous images: normal and benign, and two categories of noncancerous images: in situ carcinoma and invasive carcinoma. The training set is composed of a total of 249 images distributed as normal (55), benign (69), in situ carcinoma (63), and invasive carcinoma (62). The initial test set composed of 20 images distributed as normal (5), benign (5), in situ carcinoma (5), and invasive carcinoma (5). Furthermore, an extended test set with increased ambiguity is also provided in which a total of 16 images are available. The images of the extended test set were distributed as normal (4), benign (4), in situ carcinoma (4), and invasive carcinoma (4). These images are available on https://rdm.inesctec.pt/dataset/nis-2017-003.
- ICIAR 2018 breast histology image data set: This data set is the extended version of the Bioimaging 2015 breast histology classification challenge data set [71]. It consists of a training set of 400 images divided into normal (100), benign (100), in situ carcinoma (100), and invasive carcinoma (100). The image size and magnification is the same as the data set provided in the study by Araujo et al. [71]. The test set consists of 100 images in which the class labels were not disclosed. These images are available on https://rdm.inesctec.pt/dataset/nis-2017-003.
- Atlas of breast histopathology [82]: The researchers built a virtual microscopy framework for the automatic scanning of digital slides of breast histopathology. The slides were made available through the web (http://www.webmicroscope.net/breastatla). We can see any part of the breast tissue sample on any magnification for pathology examinations. This data set has more than 150 histopathology images of breast cancer provided by the University of Tampere, Finland.
- TMA database: The Stanford Tissue Microarray (TMA) database contains 5237 images of in total. In this data set, 5044 images represent different diseases and disorders, and 1524 images are related to breast disorder. In the images of breast disorders, most of the images were diagnosed as breast carcinoma. This data set is available on http://tma.im/cgi-bin/home.pl.

Results

The performance of many ML-based CAD systems for the diagnosis of breast cancer histopathology is listed in Tables 1 and 2 (see Appendix A). These tables are available as supplementary data. The recent research activities show that BreaKHis is the most commonly used data set by the researchers for the diagnosis of breast cancer using histopathology. For this data set, it can be seen that deep learning (CNN)–based approaches have shown the outstanding performance as compared with the traditional handcrafted feature extraction based approaches [3,34,35]. This suggests the features learned by the CNN are better as compared with the handcrafted features. The CNN-based approaches proposed in [24,41,83,84] show comparatively high performance as compared with the CNN based approaches proposed in [37,85,86]. This suggests that the transfer learning–based training strategy can effectively improve the diagnosis performance as compared with training the networks from scratch. This is because training CNN from scratch requires a large data set to enhance training performance. But, BreaKHis data set provides limited

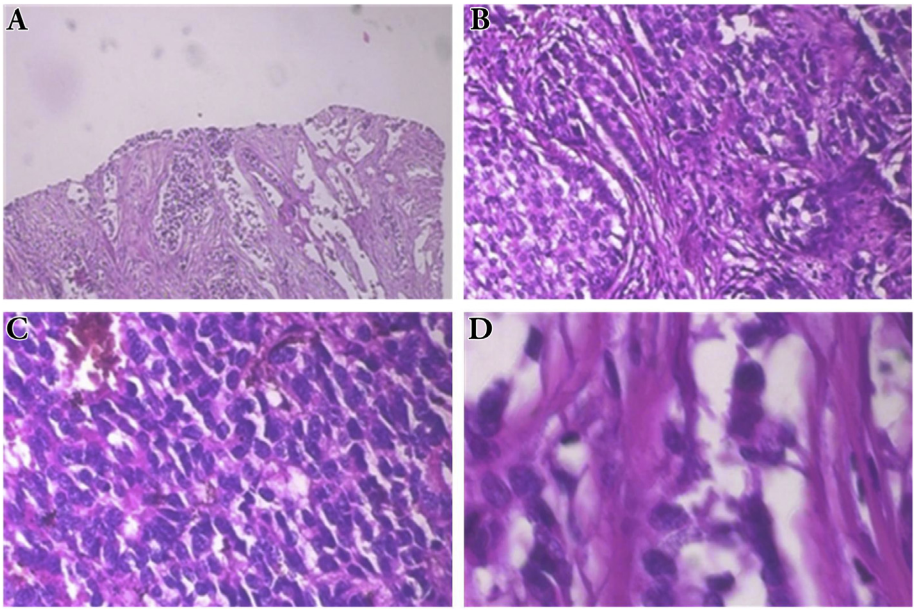

Figure 3. Digital images of BreakHis data set perceived on different magnifications (A) 40×, (B) 100×, (C) 200×, (D) 400×.

samples for training. So, the pretrained CNNs produce better results for the classification of the BreaKHis data set as compared with the CNNs trained from scratch. Furthermore, Gandomkar et al [24] have outperformed the other CNN-based approaches for the binary as well as multiclassification of the BreaKHis data set. In this study, the authors used stain normalization, oversampling, and a deep ResNet model with transfer learning to classify the images. The reported results of ML based CAD systems on the data sets other than BreaKHis are shown in Table 1. For Bioimaging 2015 breast histology classification challenge data set, the approach of Li et al [72] performs better as compared with the approach of Araujo et al [71]. In the study by Araujo et al [71], the authors divided the image into 512 ×512 size patches and extracted the features of their own designed CNN, whereas in the study by Li et al [72], the authors used comparatively small size patches and used pretrained CNN (ResNet50) to extract the features. This suggests that the features of pretrained CNN are more powerful as compared with the features of any arbitrarily designed CNN. In addition, Roy et al [87] used their own designed CNN trained from scratch for binary and multiclassification of the ICIAR 2018 breast histology image data set. However, the results of the study by Roy et al [87] cannot be compared with [71,72] because the authors used extended data (400 microscopy images) to create training, validation, and test set. Anuranjeeta et al [5] used BisQue data set segmentation, feature extraction, and classification and achieved 85.7% accuracy. Belsare et al [32] used segmentation, handcrafted feature extraction, and linear discriminant analysis classifier to achieve 100% classification accuracy. The approach of the study by Vu et al [33] gives 97.75% accuracy. However, it is not possible to compare these approaches with other research studies because the experiments were performed on the unpublished data sets.

Discussion

In this section, we discuss how conventional pathology-based diagnosis systems can become more advanced using ML methods. Furthermore, we also present potential gaps in the current ML approaches and their applicability for breast cancer diagnosis using histopathology.

Impact of Using Machine Learning for Histopathology

Pathologists recognize an image as cancerous or noncancerous by examining the images under the microscope. Histopathology images have complex visual patterns that are difficult to diagnose using the human eye. The cancer diagnosis using histopathology is a complex and time-consuming process for a pathologist. Pathological diagnosis can benefit by using ML approaches due to the factors delineated here.

- Improving the image quality: Raw histopathology images may have noisy artifacts, low pixel quality, and illumination (depending on the quality of color camera attached with a microscope), and color variations due to the differences in stain coloring. It creates difficulty in identifying the focal areas within the image. The quality of these images can be improved with the help of image preprocessing methods such as thresholding, noise removal, contrast adjustment, histogram equalization, and color normalization strategies.
- Identification of local entities: Segmentation helps to identify the local entities (cells) present in the image. These entities can be further described with the help of the morphological features such as size, shape, area, etc. Morphological features play a significant role in classifying the histopathology image as cancerous or noncancerous.

- Feature learning: Handcrafted features provide quantifiable information from raw histopathological images that are useful for cancer diagnosis. Furthermore, CNNs automatically learn the features well-suited for the determination of cancer.
- Generalization capability: The learned features from the labeled (cancer/noncancer) histopathology images can be generalized to the previously unseen histopathology images. Furthermore, histopathology images share some common features such as lines, edges, and color, with the images of the natural scene. So, the CNNs previously trained on a large number of natural scene images may be utilized for extracting the common features of histopathological images. This is very useful when we have a large number of unlabeled histopathology images and a small number of labeled training images.
- Invariant and fast diagnosis result: In a hospital, several patients are prescribed for cancer biopsy leading to massive and complicated histopathology data. The accuracy of diagnosis depends on the experience of the pathologist. Various factors such as heavy workload, fatigue, and time availability may affect the accuracy of diagnosis [47]. Because ML approaches use mathematical models to classify the images into cancer and noncancer, the diagnosis results remain unaffected by these factors. Furthermore, the ML models provide fast and precise diagnosis results for a large number of histopathology images in comparison with the human expert.

Real-World Applicability

Although ML-based CAD systems have shown promising performance for cancer diagnosis as compared with the human expert, there are some research gaps that need to be addressed for the application of these systems in a real-world environment:

- Large-scale training data set: ML (especially deep learning)-based CAD systems require massive training data to produce the correct diagnosis result for the unseen data in a real pathology environment. However, it is difficult to obtain a large-scale training data set from local hospitals due to factors such as partnerships with clinics and hospitals, image access permission, ethical issues, etc. [12]. Some researchers used small size private (unpublished) data sets to develop the CAD systems. These systems may not produce the correct result when deployed in the real clinical environment. In addition, the available public histopathology data sets are small and do not meet the requirements of real-world clinical pathology applications. For example, the BisQue data set consists of only 32 benign and 26 malignant images. Bioimaging 2015 breast histology classification challenge data set has on an average of 62 training images of each category. ICIAR 2018 breast histology image data set has 100 images per category that needs to be distributed into the training and test set. Among all the available public data sets of breast cancer histopathology, the size of the BreaKHis data set is largest. The data set may be suitable for image-level diagnosis. But, for the patient-level diagnosis, the data set composed of the images corresponding to only 82 patients (25 benign and 57 malignant) divided into eight subcategories, that is, on an average of 10 patients per category.
- Localization of area of interest in histopathology images: Typically, in the clinical practice, only a few small areas within the histopathological images are used by pathologists to classify the image into the cancer or noncancer category. In publicly available microscopy data sets of histopathology, such areas are neither segmented nor labeled, that is, the exact location of cancerous and noncancerous regions within the image is unknown.
- Class imbalance problem: Among all the data sets, BreaKHis is the most commonly used data set used by the researchers for computer-aided breast cancer diagnosis using histopathology. Moreover, BreaKHis is an imbalanced data set containing malignant images in the majority as compared with the benign class. The class imbalance ratio increases when the image data set is used for multiclassification. Consequently, for a regular ML algorithm, the benign class has fewer instances for training for this data set. So, the benign class has comparatively a higher probability of misclassification, and the performance is biased toward the malignant class. Moreover, the reliability of a CAD system vanishes if it recognizes a benign patient as malignant. It may create a situation to make a wrong diagnostic decision for the pathologist. Most of the research work [3,35,37,40,41] did not use any strategy to handle the minority class instances and measured the performance of their classifiers in terms of the overall Prrs and average image classification accuracy. In the case of an imbalanced data set such as BreaKHis, providing the image classification accuracy and Prr is not sufficient to describe the performance of any classifier. The methods used to deal with imbalanced data set in ML are oversampling, undersampling, and algorithm level methods [88]. However, a limited investigation has been found on the application of these methods for improving the minority class performance in the classification of breast cancer histopathology images. For example, the authors in [24] used oversampling to enhance the minority class instances. The authors extracted 20 patches corresponding to the minority (benign) class images, whereas for the majority class images, the authors extracted only ten patches. But, this method may fail to distinguish between the informative and redundant patches and increase the training overhead. In addition, the neighborhood of the image patches may not necessarily share the same label because the malignant benign and malignant images may have some disease-free regions within the image.

Conclusion

In this review, we have detailed the application of ML methods for computer-aided breast cancer diagnosis using conventional photomicroscopy-based histopathology images. ML-based CAD systems have shown remarkable advancement in traditional microscopy-based diagnosis systems. These modalities have gained the interest of researchers because of the publicly available histopathology data sets and the ability to extract powerful features from the histopathology images. Conventional handcrafted feature extraction approaches have been outperformed by deep learning approaches because of their feature learning and generalization capabilities. In spite of the success of developing fast and accurate CAD systems, it is unclear whether these systems are suitable for real-world applications of clinical pathology. This could be challenging due to the lack of a real-time large-scale data set for training these systems and a class imbalance problem.

ML research for computer-aided breast cancer diagnosis is still hopeful, we see the following advancements in future:

- Large-scale cancer diagnosis: Although breast cancer diagnosis on large-scale histopathology data set has not been explored by the researchers, deep learning models have the necessary potential to learn from a large-scale data set and generalize the features to the unseen data in real pathology environment.
- Localization: In the future, deep learning models can be designed to localize the cancerous and noncancerous regions within the image.
- Robust to noise: Given the histopathology images of a real-world environment, deep learning methods can learn to remove the noisy artifacts while learning the quality features for the diagnosis.
- Reliability: For ML-based CAD systems to be reliable in a real pathology environment, they must be incorporated with class imbalanced learning methods; for example, assigning more weights to the minority class images as compared with the majority class images while training the ML model so that the learning will not be biased toward the majority class.
- Developing pretrained CNN for histopathology: Current research is focused on reusing the CNNs previously trained on ImageNet data for extracting the features of histopathology images. In the future, a CNN trained on large-scale histopathology images can be made available for the feature extraction and transfer learning tasks specific to breast cancer and the other types of cancer.
- Fusion of multiple modalities: The cost of implementing CNNs is high because of the increased memory requirement and long training time. In the future, the fully connected neural network layers of CNN can be replaced with a fast classifier such as extreme learning machines to speed up the training process.

Acknowledgements

The authors acknowledge Dr. Neelkamal Kapoor and Dr. Deepti Joshi from the All India Institute of Medical Sciences, Bhopal, to help them in understanding the histopathological data.

Supplementary Data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jmir.2019.11.001.

References

[1] Street, W. N. (1994). Cancer Diagnosis and Prognosis via Linear –Programming-Based Machine Learning. Ph. D. Dissertation. Madison, WI: The University of Wisconsin- Madison. UMI Order No. GAX94-32979.

[2] Jalalian, A., Mashohor, S., Mahmud, R., Karasfi, B., Saripan, M. I., & Ramli, A. R. (2017). Review article: Foundation and methodologies in computer-aided diagnosis systems for breast cancer detection. *EXCLI J* 16, 113–137.

[3] Spanhol, F. A., Oliveira, L. S., Petitjean, C., & Heutte, L. (2016). A dataset for breast cancer histopathological image classification. *IEEE Trans Biomed Eng* 63(7), 1455–1462.

[4] Arevalo, J., & Cruz-roa, A. (2014). Histopathology image representation for autometic analysis: a state–of—the-art review. *Rev Med* 22(2), 79–91.

[5] Anuranjeeta, Shukla, K. K., Tiwari, A., & Sharma, S. (2017). Classification of histopathological images of breast cancerous and non cancerous cells based on morphological features. *Biomed Pharmacol J* 10(1), 353–366.

[6] Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. Proceedings of the International Society for Optical Engineering (SPIE) 1905, Biomedical Image Processing and Biomedical Visualization, vol. 1905, Acharya, R. S. and Goldgof, D. B., Eds (pp. 861–870).

[7] Dua, D., & Graff, C. (2019). UCI Machine Learning Repository. Irvine, CA: University of California, School of Information and Computer Science. Available at: http://archive.ics.uci.edu/ml. Accessed December 11, 2019.

[8] Cruz, J. A., & Wishart, D. S. (2006). Applications of machine learning in cancer prediction and prognosis. *Cancer Inform* 2, 59–77.

[9] Kourou, K., Exarchos, T. P., Exarchos, K. P., Karamouzis, M. V., & Fotiadis, D. I. (2015). Machine learning applications in cancer prognosis and prediction. *Comput Struct Biotechnol J* 13, 8–17.

[10] Komura, D., & Ishikawa, S. (2018). Machine learning methods for histopathological image analysis. *Comput Struct Biotechnol J* 16, 34–42.

[11] Hu, Z., Tang, J., Wang, Z., Zhang, K., Zhang, L., & Sun, Q. (2018). Deep learning for image-based cancer detection and diagnosis–a survey. *Pattern Recognit* 83, 134–149.

[12] Gonçalves, V. M., Delamaro, M. E., & De Lourdes, F. (2014). A systematic review on the evaluation and characteristics of computer- aided diagnosis systems. *Braz J Biomed Eng* 30(4), 355–383.

[13] Gandomkar, Z., Brennan, P. C., & Mello-Thoms, C. (2016). Computer-based image analysis in breast pathology. *J Pathol Inform* 7(1), 43.

[14] Aswathy, M. A., & Jagannath, M. (2017). Detection of breast cancer on digital histopathology images: present status and future possibilities. *Inform Med Unlocked* 8, 74–79.

[15] Bakator, M., & Radosav, D. (2018). Deep learning and medical diagnosis: a review of literature. *Multimodal Technol Interact* 2(3), 47.

[16] Evans, A. J., Salama, M. E., Henricks, W. H., & Pantanowitz, L. (2017). Implementation of whole slide imaging for clinical purposes issues to consider from the perspective of early adopters. *Arch Pathol Lab Med* 141(7), 944–959.

[17] He, L., Long, L. R., Antani, S., & Thoma, G. R. (2012). Histology image analysis for carcinoma detection and grading. *Comput Methods Programs Biomed* 107(3), 538–556.
[18] Sembiring, R. W., & Mohamad Zain, J. (2011). The design of preprocessing multidimensional data based on component analysis. *Comput Inf Sci* 4(3), 106–115.
[19] Demir, C., & Yener, B. (2005). Automated cancer diagnosis based on histopathological images: a systematic survey. New York, NY: Rensselaer Polytechnic Institute Technical Report TR-05-09.
[20] Zuiderveld, K. (1994). Contrast limited adaptive histogram equalization. In: *Graphics gems IV* (pp. 474–485). San Diego, CA: Academic Press Professional.
[21] Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Trans Syst Man Cybern* 9(1), 62–66.
[22] Tellez, D., Litjens, G., & Bandi, P., et al. (2019). Quantifying the effects of data augmentation and stain color normalization in convolutional neural networks for computational pathology. *Med Image Anal* 58, 101544.
[23] Ciompi, F., Geessink, O., & Bejnordi, B. E., et al. (2017). The importance of stain normalization in colorectal tissue classification with convolutional networks. 2017 IEEE 14th International Symposium on Biomedical Imaging (ISBI 2017) (pp. 160–163).
[24] Gandomkar, Z., Brennan, P. C., & Mello-Thoms, C. (2018). MuDeRN: multi-category classification of breast histopathological image using deep residual networks. *Artif Intell Med* 88, 14–24.
[25] Vo, D. M., Nguyen, N. Q., & Lee, S. W. (2019). Classification of breast cancer histology images using incremental boosting convolution networks. *Inf Sci* 482, 123–138.
[26] Roy, S., kumar Jain, A., Lal, S., & Kini, J. (2018). A study about color normalization methods for histopathology images. *Micron* 114, 42–61.
[27] Reinhard, E., Ashikhmin, M., Gooch, B., & Shirley, P. (2001). Color transfer between images. *IEEE Comput Graph Appl* 21(5), 34–41.
[28] Macenko, M., Niethammer, M., & Marron, J. S., et al. (2009). A method for normalizing histology slides for quantitative analysis. Proceedings–2009 IEEE International Symposium on Biomedical Imaging: From Nano to Macro, ISBI 2009 (pp. 1107–1110).
[29] Khan, A. M., Rajpoot, N., Treanor, D., & Magee, D. (2014). A nonlinear mapping approach to stain normalization in digital histopathology images using image-specific color deconvolution. *IEEE Trans Biomed Eng* 61(6), 1729–1738.
[30] Kotsiantis, S. B. (2007). Supervised machine learning : a review of classification techniques. *Informatica* 31(3), 249–268.
[31] Kumar, G., & Bhatia, P. K. (2014). A detailed review of feature extraction in image processing systems. International Conference on Advanced Computing and Communication Technologies, ACCT (pp. 5–12).
[32] Belsare, A. D., Mushrif, M. M., Pangarkar, M. A., & Meshram, N. (2015). Classification of breast cancer histopathology images using texture feature analysis. Proceedings of the 2015 IEEE Region 10 Conference (TENCON 2015),Macao, China (pp. 1–5).
[33] Vu, T. H., Mousavi, H. S., Monga, V., Rao, G., & Rao, U. K. A. (2016). Histopathological image classification using discriminative feature-oriented dictionary learning. *IEEE Trans Med Imaging* 35(3), 738–751.
[34] Gupta, V., & Bhavsar, A. (2017). Breast cancer histopathological image classification: is magnification important?. 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW) (pp. 769–776).
[35] Sudharshan, P. J., Petitjean, C., Spanhol, F., Oliveira, L. E., Heutte, L., & Honeine, P. (2019). Multiple instance learning for histopathological breast cancer image classification. *Expert Syst Appl* 117, 103–111.
[36] Cruz-Roa, A., Basavanhally, A., & González, F., et al. (2014). Automatic detection of invasive ductal carcinoma in whole slide images with convolutional neural networks. Proceedings of the International Society for Optical Engineering (SPIE) 9041, Medical Imaging 2014: Digital Pathology, vol. 9041, no. 216 (pp. 904103).
[37] Spanhol, F. A., Oliveira, L. S., Petitjean, C., & Heutte, L. (2016). Breast cancer histopathological image classification using convolutional neural networks. Neural Networks (IJCNN), 2016 International Joint Conference on, 2016 (pp. 2560–2567).
[38] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, Kai., & Fei-Fei, Li (2009). ImageNet: a large-scale hierarchical image database. 2009 IEEE Conference on Computer Vision and Pattern Recognition (pp. 248–255).
[39] Shallu, & Mehra, R. (2018). Breast cancer histology images classification: Training from scratch or transfer learning? *ICT Express* 4(4), 247–254.
[40] Spanhol, F. A., Oliveira, L. S., Cavalin, P. R., Petitjean, C., & Heutte, L. (2017). Deep features for breast cancer histopathological image classification. 2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC) (pp. 1868–1873).
[41] Han, Z., Wei, B., Zheng, Y., Yin, Y., Li, K., & Li, S. (2017). Breast cancer multi-classification from histopathological images with structured deep learning model. *Sci Rep* 7(1), 1–10.
[42] Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. *Adv Neural Inf Process Syst* 1, 1–9.
[43] Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. *Inf Softw Technol* 51(4), 769–784.
[44] Szegedy, C., Liu, W., & Jia, Y., et al. (2015). Going deeper with convolutions. IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (pp. 1–9).
[45] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proc IEEE* 1–9.
[46] Hira, Z. M., & Gillies, D. F. (2015). A review of feature selection and feature extraction methods applied on microarray data. *Adv Bioinformatics* 2015, 198363.
[47] Gurcan, M. N., Boucheron, L. E., Can, A., Madabhushi, A., Rajpoot, N. M., & Yener, B. (2009). Histopathological image analysis: a review. *IEEE Rev Biomed Eng* 2, 147–171.
[48] Mladenic, D. (2006). Feature selection for dimensionality reduction. In:Saunders, C, Grobelnik, M., Gunn, S., & Shawe-Taylor, J. (Eds.). (2006). *Subspace, Latent Structure and Feature Selection* vol. 3940 (pp. 84–102). Berlin, Heidelberg: Springer-Verlag.
[49] Pudil, P., Novovičová, J., & Kittler, J. (1994). Floating search methods in feature selection. *Pattern Recognit Lett* 15(11), 1119–1125.
[50] Jain, A., & Zongker, D. (1997). Feature selection: evaluation, application, and small sample performance. *IEEE Trans Pattern Anal Mach Intell* 19(2), 153–158.
[51] Freund, Y., & Schapire, R. E. (1997). A decision-theoretic generalization of on-line learning and an application to boosting. *J Comput Syst Sci* 55(1), 119–139.
[52] Perkins, S., Lacker, K., & Theiler, J. (2003). "Grafting : fast, incremental feature selection by gradient descent in function space. *J Mach Learn Res* 3, 1333–1356.
[53] Xue, B., Zhang, M., Member, S., & Browne, W. N. (2013). Particle swarm optimization for feature selection in classification : a multi-objective approach. *IEEE Trans Cybern* 43(6), 1656–1671.
[54] Scherer, D., Müller, A., & Behnke, S. (2010). "Evaluation of pooling operations in convolutional architectures for object recognition. In:Diamantaras, K, Duch, W., & Iliadis, L. S. (Eds.). (2010). *Artificial Neural Networks – ICANN 2010. ICANN 2010 (Conference). Lecture Notes in Computer Science* vol. 6354 (pp. 92–101). Berlin, Heidelberg: Springer.
[55] Hinton, G. E., Srivastava, N., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. R. (2012). Improving neural networks by preventing co-adaptation of feature detectors. arXiv: 1207.0580 [cs.NE].
[56] Ng, A. Y. (2004). Feature selection, L1 vs. L2. Twenty-first Int. Conf. Mach. Learn. - ICML '04 (pp. 78).
[57] Hastie, T., Tibshirani, R., & Friedman, J. (2001). Basis expansions and regularization. In: *The Elements of Statistical Learning* (pp. 139–189). New York, NY: Springer.

[58] Serpen, G., & Pathical, S. (2009). Classification in high-dimensional feature spaces: random subsample ensemble. Proceeding of the International Conference on Machine Learning and Applications (ICMLA'09). Miami Beach, Florida, USA (pp. 740–745).
[59] Min, S.-H. (2016). A genetic algorithm-based heterogeneous random subspace ensemble model for bankruptcy prediction. *Int J Appl Eng Res* 11(4), 2927–2931.
[60] Witten, I. H., & Frank, E. (2005). Data mining practical machine learning tools and techniques (2nd ed.). Morgan Kaufmann Publishers.
[61] Amin, R. K., Indwiarti, & Sibaroni, Y. (2015). Implementation of decision tree using C4 . 5 algorithm in decision making of loan application by debtor (Case study : Bank pasar of yogyakarta special region). Proceeding of the 3rd International Conference on Information and Communication Technology (ICoICT) (pp. 75–80).
[62] Lim, T., Loh, W.-Y., & Shih, Y.-S. (2000). A comparison of prediction accuracy, complexity, and training time of thirty-three old and new classification algorithms. *Mach Learn* 40(3), 203–228.
[63] Ruggieri, S. (2002). Efficient C4 .5 [classification algorithm]. *IEEE Trans Knowl Data Eng* 14(2), 438–444.
[64] Ripley, R. M. (1998). Neural network models for breast cancer prognosis. Ph.D. Dissertation. St. Cross College: University of Oxford, UK.
[65] Rani, K. U. (2010). Parallel approach for diagnosis of breast cancer using neural network technique. *Int J Comput Appl* 10(3), 1–5.
[66] Jha, G. K. (2007). Artificial neural networks and its applications. New Delhi, India: Indian Agricultural Statistics Research Institute (I.C.A.R.).
[67] Paulin, F., & Santhakumaran, A. (2011). Classification of breast cancer by comparing back propagation training algorithms. *Int J Comput Sci Eng* 3(1), 327–332.
[68] Salama, G. I., Abdelhalim, M. B., & Zeid, M. A. (2012). Breast cancer diagnosis on three different datasets using multi-classifiers. *Int J Comput Inf Technol* 1(1), 36–43.
[69] Introduction to Support Vector Machines-OpenCV 2.4.13.6 documentation. (2018). Available at: https://docs.opencv.org/2.4/doc/tutorials/ml/introduction_to_svm/introduction_to_svm.html. Accessed April 23, 2018.
[70] Osareh, A., & Shadgar, B. (2010). Machine learning techniques to diagnose breast cancer. 5th International Symposium on Health Informatics and Bioinformatics, 2010 (pp. 114–120).
[71] Araujo, T., Aresta, G., & Castro, E., et al. (2017). Classification of breast cancer histology images using convolutional neural networks. *PLoS One* 12(6), 1–14.
[72] Li, Y., Wu, J., & Wu, Q. (2019). Classification of breast cancer histology images using multi-size and discriminative patches based on deep learning. *IEEE Access* 7, 21400–21408.
[73] Breiman, L. (1996). Bagging predictors. *Mach Learn* 24, 123–140.
[74] Freund, Y., & Schapire, R. E. (1996). Experiments with a new boosting algorithm. In L. Saitta (Ed.), *Proceedings of the Thirteenth International Conference on International Conference on Machine Learning (ICML'96)* (pp. 148–156). San Francisco, CA: Morgan Kaufmann Publishers Inc..
[75] Ho, T. K. (1998). The random subspace method for constructing decision forests. *IEEE Trans Pattern Anal Mach Intell* 20(8), 832–844.
[76] Dietterich, T. G. (1990). Ensemble methods in machine learning. In M. A. Arbib (Ed.), *The handbook of brain theory and neural networks* (2nd ed.) (pp. 406–408) Massachusetts London, England: Cambridge: MIT Press.
[77] Alom, M. Z., Taha, T. M., & Yakopcic, C., et al. (2018). The history began from AlexNet: a comprehensive survey on deep learning approaches. arXiv: 1803.01164 [cs.CV].
[78] Prasoon, A., Petersen, K., Igel, C., Lauze, F., Dam, E., & Nielsen, M. (2013). Deep feature learning for knee cartilage segmentation using a triplanar convolutional neural network. In:Mori, K, Sakuma, I., Sato, Y., Barillot, C., & Navab, N. (Eds.). (2013). *Medical image computing and computer-assisted intervention—MICCAI 2013. Lecture notes in computer science* vol. 8150 (pp. 246–253). Berlin, Heidelberg: Springer.
[79] Ciresan, D. C., Giusti, A., Gambardella, L. M., & Schmidhuber, J. (2012). Deep neural networks segment neuronal membranes in electron microscopy images. Advances in Neural Information Processing Systems 25, proceedings of the Neural Information Processing Systems Conference (NIPS 2012), Lake Tahoe, Nevada, USA (pp. 1–9).
[80] LeCun, Y., Kavukcuoglu, K., & Farabet, C. (2010). Convolutional networks and applications in vision. Proceedings of 2010 IEEE International Symposium on Circuits and Systems (pp. 253–256).
[81] Drelie Gelasca E., Byun J., Obara B., & Manjunath B. S., Evaluation and benchmark for biological image segmentation in IEEE International Conference on Image Processing, 1816–1819.
[82] Lundin, M., Lundin, J., Helin, H., & Isola, J. (2004). A digital atlas of breast histopathology: an application of web based virtual microscopy. *J Clin Pathol* 57(12), 1288–1291.
[83] Nahid, A., & Kong, Y. (2018). Histopathological breast-image classification using local and frequency domains by convolutional neural network. *Information* 9(1), 1–26.
[84] Xie, J., Liu, R., Luttrell, J., & Zhang, C. (2019). Deep learning based analysis of histopathological images of breast cancer. *Front Genet* 10, 80.
[85] Bayramoglu, N., Kannala, J., & Heikkila, J. (2016). Deep learning for magnification independent breast cancer histopathology image classification. In: *Proceedings of the 23rd International Conference on Pattern Recognition (ICPR)* (pp. 2440–2445). Mexico: Cancun.
[86] Reza, M. S., & Ma, J. (2018). Imbalanced histopathological breast cancer image classification with convolutional neural network. 2018 14th IEEE International Conference on Signal Processing (ICSP), vol. 2018-Augus (pp. 619–624).
[87] Roy, K., Banik, D., Bhattacharjee, D., & Nasipuri, M. (2019). Patch-based system for Classification of Breast Histology images using deep learning. *Comput Med Imaging Graph* 71, 90–103.
[88] Haibo, H., & Garcia, E. A. (2009). Learning from Imbalanced Data. *IEEE Trans Knowl Data Eng* 21(9), 1263–1284.